

Supplementary Material: Datasets and Data Preprocessing in Machine Learning-Based Insider Threat Detection: A Systematic Literature Review

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Note: This Supplementary material can also be accessed online on GitHub at:
<https://github.com/alriyami7777-alt/SLR-on-Datasets-and-Data-Preprocessing--Supplement-Alriyami-Murtadha>

1. Overview

This repository provides the supplementary material and audit trail for the SLR titled “Datasets and Data Preprocessing in Machine Learning-Based Insider Threat Detection: A Systematic Literature Review.” It supports transparency and reproducibility by documenting the search strategy, PRISMA 2020 study selection process, quality assessment framework, data extraction records, and visualization datasets used in the review of 110 primary studies.

AI-assisted tools were used only to support initial extraction, screening organization, and visualization preparation. All study inclusion decisions, extracted fields, classifications, counts, and final interpretations were reviewed and verified by the authors.

2. Repository Structure

The repository is organized into six folders:

Folder Name	Primary Content	Format
01_Search_Strategy	Boolean search strings, database search logs, and raw record counts. RIS and CSV files from the databases	PDF RIS CSV
02_Study_Selection	The multi-stage screening and selection process and the PRISMA 2020 flow diagram	PDF / CSV
03_Quality_Assessment	Detailed QA1–QA5 scoring for every paper and the Exclusion Log for the 7 papers excluded during full-text eligibility	PDF / Excel
04_Data_Extraction	Structured data extraction tables mapped directly to RQ1–RQ4 .	PDF
05_Visualizations	Bibliometric data from publication trends and avenues, and keyword frequency visualization.	PDF / Excel
06_PRISMA_Checklists	Completed PRISMA 2020 checklist and PRISMA 2020 checklist for Abstract.	PDF

3. Methodology & Transparency

To improve transparency and reproducibility, this repository documents two supporting workflows used in the review:

- **AI-Assisted Extraction (Folder 04):** NotebookLM was used to support the initial extraction of technical parameters from the 110 included primary studies. All extracted information, classifications, and interpretations were reviewed and verified by the authors.
- **Visualizations (Folder 05):** Visualizations were prepared using a hybrid workflow that combined qualitative thematic scanning, quantitative data synthesis, Google Colab,

Microsoft Excel, and Google Gemini for graphical support. Final figures, counts, and labels were manually checked.

4. How to Use This Repository

01_Search_Strategy: For Replication & Verification

- **Replicate Searches:** Use the provided Boolean search strings to run comparable searches across Web of Science, ScienceDirect, ACM Digital Library, IEEE Xplore, and Scopus.
- **Audit Raw Data:** The .ris and .csv files contain the raw database records before deduplication, showing what each database returned on the search date.

02_Study_Selection: For Procedural Transparency

- **Track the Process:** The PRISMA 2020 flow diagram provides a visual summary of the identification, screening, eligibility, and inclusion stages.

03_Quality_Assessment: For Rigor Validation

- **Review Scoring:** Provides details on how the included studies were assessed against the five Quality Assessment (QA) criteria.
- **Verify Exclusions:** It lists the 7 full-text papers that were excluded and provides the specific reasons for their removal.

04_Data_Extraction: For Technical Synthesis

- **Answer Research Questions:** The structured extraction data are mapped to RQ1–RQ4 and can be used to locate information on dataset types, dataset characteristics, preprocessing techniques, and data-related challenges.
- **AI Methodology:** The AI-assisted extraction document explains how LLM-supported extraction was used to accelerate synthesis while maintaining human-in-the-loop verification.

05_Visualizations: For Trend Analysis

- **Keyword & Bibliometrics:** The frequency datasets can be used to examine publication patterns, keyword trends, and recurring themes in the reviewed literature.
- **Identify Leading Venues:** The venue data show which journals published the most relevant studies in this review.

06_PRISMA_Checklists: For Standard Compliance

- **Verify Compliance:** These PDF files show where the main manuscript addresses the required PRISMA 2020 reporting items.